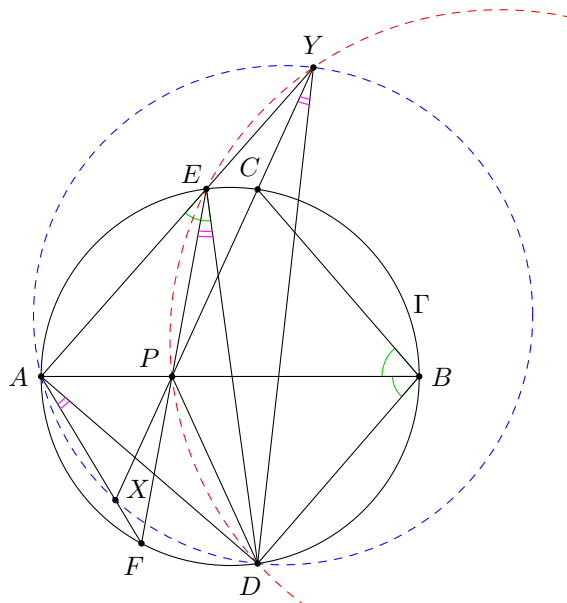


Problem 9 (Czech-Polish-Slovak Match 2019 Problem 1). Let AB be a diameter of a circle Γ . Another circle with centre B intersects Γ at C and D , and intersects the diameter AB at P . A line passing through P meets Γ at E and F such that D, F, A, E, C lie in this order on Γ . The line CP intersects the segment AF at X , and intersects the ray AE at Y . Prove that A, X, D, Y are concyclic.



Solution. By symmetry, AB bisects $\angle CBD$. Since B is the centre of (CDP) , we have

$$\angle DPC = \frac{1}{2}(360^\circ - \angle CBD) = 180^\circ - \angle ABD = 180^\circ - \angle AED = \angle DEY.$$

This implies E, P, D, Y are concyclic. Then we have

$$\angle XAD = \angle FAD = \angle FED = \angle PED = \angle PYD = \angle XYD.$$

Thus, A, X, D, Y are concyclic. □

Problem 10 (Japan MO 2016 Problem 3). Let n be a positive integer. In a country, there is a king and 2^n citizens. Every citizen has an infinite number of banknotes and a finite number of coins. The value of a banknote is $\$2^n$, and the value of a coin can be one of $\$1, \$2, \$2^2, \dots, \2^{n-1} . Let S be the total number of coins in the country. The following policy is implemented.

- Every day, each citizen must give a certain amount of money to either another citizen or the king using the banknotes and coins that the citizen has.
- The amount of money given by each citizen is $\$1$ more than the total amount of money he/she receives on that day.
- The citizens can only give away money that they currently have. Money received on that day cannot be used immediately. Also, no change is allowed.

Find the smallest possible value of S such that this policy can be implemented indefinitely (given that all citizens are cooperative).

Solution. The answer is $n \cdot 2^{n-1}$.

We first show that S can be $n \cdot 2^{n-1}$. Suppose the values of the coins owned by the citizens are $0, 1, \dots, 2^n - 1$ respectively. For convenience, we label these citizens by $0, 1, \dots, 2^n - 1$. For $k = 1, 2, \dots, 2^n - 2$, citizen k gives all his coins to citizen $k + 1$. Citizen $2^n - 1$ gives all his coins to citizen 0. Citizen 0 gives a banknote to the king. All conditions are satisfied. Afterwards, the coins owned by the citizens are just shifted in a cyclic way. Therefore, this can proceed indefinitely. We count the smallest number

of coins in total. To get a total value of k , there should be at least $f(k)$ coins, where $f(k)$ denotes the number of 1's in the binary representation of k . Thus, we have

$$S \geq f(0) + f(1) + \cdots + f(2^n - 1),$$

which is the total number of 1's in the binary representations of $0, 1, \dots, 2^n - 1$. Note that all these binary representations have n digits (possibly starting with 0). For each of the n digits, it is equal to 1 for exactly 2^{n-1} numbers since there are 2 choices for each of the remaining digits. Thus, we have

$$S \geq f(0) + f(1) + \cdots + f(2^n - 1) = n \cdot 2^{n-1}.$$

Next, we show that S is at least $n \cdot 2^{n-1}$. The value of the coins owned by each citizen modulo 2^n is decreased by 1 every day. Therefore, for any 2^n consecutive days, this value runs through each of $0, 1, \dots, 2^n - 1$ once. Thus, each citizen owns at least

$$f(0) + f(1) + \cdots + f(2^n - 1) = n \cdot 2^{n-1}$$

coins in total. As there are 2^n citizens, on average the total number of coins on each day is at least $\frac{2^n \cdot n \cdot 2^{n-1}}{2^n} = n \cdot 2^{n-1}$. Thus, there is a day for which the total number of coins is at least $n \cdot 2^{n-1}$. Since the total number of coins is decreasing, this implies $S \geq n \cdot 2^{n-1}$ and we are done. $\square$

Problem 11 (Spain MO 2018 Problem 6). Find all function(s) $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $f(x + f(y)) = yf(xy + 1)$ for all $x, y \in \mathbb{R}^+$.

Solution. Label the equation as follows.

$$f(x + f(y)) = yf(xy + 1) \tag{1}$$

If $y \neq 1$ and $\frac{f(y) - 1}{y - 1} > 0$, then we can put $x = \frac{f(y) - 1}{y - 1}$ in (1) so that $x + f(y) = xy + 1$. This gives $y = 1$, contradicting the assumption $y \neq 1$. Thus, for any $y \neq 1$, we must have $\frac{f(y) - 1}{y - 1} \leq 0$. In other words, if $y > 1$, then $f(y) \leq 1$, while if $y < 1$, then $f(y) \geq 1$.

For any $y > 1$, we replace x by $\frac{y-1}{y}$ in (1) to obtain

$$f\left(1 - \frac{1}{y} + f(y)\right) = yf(y).$$

- If $f(y) > \frac{1}{y}$, then $1 - \frac{1}{y} + f(y) > 1$, and so $f\left(1 - \frac{1}{y} + f(y)\right) \leq 1$. However, this contradicts $yf(y) > y \cdot \frac{1}{y} = 1$.
- If $f(y) < \frac{1}{y}$, then $1 - \frac{1}{y} + f(y) < 1$, and so $f\left(1 - \frac{1}{y} + f(y)\right) \geq 1$. However, this contradicts $yf(y) < y \cdot \frac{1}{y} = 1$.

It follows that $f(y) = \frac{1}{y}$ for any $y > 1$.

Next, for any $x \leq 1$, we replace x by $\frac{x}{2}$ and put $y = \frac{2}{x}$ in (1). Since $y > 1$ and $xy + 1 > 1$, this gives $f\left(\frac{x}{2} + \frac{x}{2}\right) = \frac{2}{x}f(2)$, and hence $f(x) = \frac{1}{x}$. So this holds for any $x \in \mathbb{R}^+$. The checking is trivial. $\square$

Problem 12 (Brazil MO 2017 Problem 6). Let n be a positive integer. Suppose p is a prime divisor of $n^3 - 3n + 1$. Prove that $p = 3$ or $p \equiv \pm 1 \pmod{9}$.

Solution 1. We use the following well-known result.

(Thue's lemma) For any prime number p and any integer n not divisible by p , there exist integers x and y such that $0 < |x|, |y| < \sqrt{p}$ and $xn \equiv y \pmod{p}$.

Proof. Let $k = \lfloor \sqrt{p} \rfloor + 1$. Consider all pairs (a, b) of integers such that $0 \leq a, b \leq k - 1$. Since there are $k^2 > p$ pairs in total, there must be two distinct pairs (a, b) and (c, d) such that $an - b \equiv cn - d \pmod{p}$. It follows that $(a - c)n \equiv b - d \pmod{p}$, where $0 < |a - c|, |b - d| \leq k - 1 < \sqrt{p}$. $\square$

Back to our problem. Suppose on the contrary that there exists a prime $p \equiv 2, 4, 5, 7 \pmod{9}$ that divides some $n^3 - 3n + 1$. We choose p to be the smallest counterexample. Clearly $p \neq 2$ and $p \nmid n$. Therefore, we can choose integers x and y such that $0 < |x|, |y| < \sqrt{p}$ and $xn \equiv y \pmod{p}$ by Thue's lemma. WLOG assume $\gcd(x, y) = 1$. Thus, we have

$$0 \equiv x^3(n^3 - 3n + 1) \equiv y^3 - 3x^2y + x^3 \pmod{p}.$$

If $3 \mid x$ or $\frac{y}{x} \equiv 0, 1 \pmod{3}$, we check that $y^3 - 3x^2y + x^3 \equiv \pm 1 \pmod{9}$, and so $\frac{y^3 - 3x^2y + x^3}{p} \equiv 2, 4, 5, 7 \pmod{9}$. If $\frac{y}{x} \equiv 2 \pmod{3}$, then

$$\frac{y^3 - 3x^2y + x^3}{3} = \frac{(3k + 2x)^3 - 3x^2(3k + 2x) + x^3}{3} \equiv \pm 1 \pmod{9},$$

so $\frac{y^3 - 3x^2y + x^3}{3p} \equiv 2, 4, 5, 7 \pmod{9}$. Therefore, $y^3 - 3x^2y + x^3$ is nonzero, and it must have an extra prime divisor q such that $q \equiv 2, 4, 5, 7 \pmod{9}$. Note that $q \nmid x$ since otherwise $q \mid x, y$, but $\gcd(x, y) = 1$. Thus, there exists an integer m such that $xm \equiv y \pmod{q}$, and we have $q \mid m^3 - 3m + 1$. By the choice of p , we must have $q \geq p$. Thus,

$$|y^3 - 3x^2y + x^3| \geq pq \geq p^2.$$

On the other hand, recall that $0 < |x|, |y| < \sqrt{p}$. Let $M = \lfloor \sqrt{p} \rfloor$. We claim that

$$|y^3 - 3x^2y + x^3| \leq 3M^3.$$

If x and y are of the same sign, then this is clear since

$$|y^3 - 3x^2y + x^3| \leq \max\{|y^3 + x^3|, |3x^2y|\} \leq 3M^3.$$

If x and y are of different signs, WLOG assume $x > 0 > y$. The result is obvious if $y^3 - 3x^2y + x^3 < 0$, since in that case

$$|y^3 - 3x^2y + x^3| = -(y^3 - 3x^2y + x^3) < -y^3 \leq M^3.$$

It remains to consider the case $x > 0 > y$ and $y^3 - 3x^2y + x^3 > 0$. Clearly, we have $y^3 - 3x^2y + x^3 \leq y^3 - 3M^2y + M^3$. This is bounded above by $3M^3$ if and only if $(M + y)^2(2M - y) \geq 0$, which is obvious. Thus, our claim holds. But then we have obtained

$$p^2 \leq |y^3 - 3x^2y + x^3| \leq 3M^3 < 3p^{\frac{3}{2}}.$$

This implies $p < 9$, and so $p = 5, 7$. It is routine to check that $n^3 - 3n + 1 \not\equiv 0 \pmod{5}$ and $n^3 - 3n + 1 \not\equiv 0 \pmod{7}$ for any integer n . So this is a contradiction, and we are done. $\square$

Solution 2. If $n \equiv t + t^{-1} \pmod{p}$ for some integer t not divisible by p , then we have

$$0 \equiv n^3 - 3n + 1 \equiv (t + t^{-1})^3 - 3(t + t^{-1}) + 1 \equiv t^3 + 1 + t^{-3} \pmod{p}.$$

This implies $t^6 + t^3 + 1 \equiv 0 \pmod{p}$, and hence $t^9 \equiv 1 \pmod{p}$. Let d be the order of t modulo p . If $d = 1, 3$, then we have $t^3 \equiv 1 \pmod{p}$, and so $0 \equiv t^6 + t^3 + 1 \equiv 3 \pmod{p}$. This implies $p = 3$. If $d = 9$, then $9 \mid p - 1$ by the Fermat little theorem.

Now, if $n \not\equiv t + t^{-1} \pmod{p}$ for every integer t not divisible by p , we consider the set

$$F = \{x + y\alpha : x, y = 0, 1, \dots, p - 1\}$$

where α satisfies the relation $\alpha^2 = n\alpha - 1$. The addition and multiplication for the elements in F are well-defined. Indeed, we have

$$\begin{aligned} (x + y\alpha) + (x' + y'\alpha) &\equiv (x + x') + (y + y')\alpha \pmod{p}, \\ (x + y\alpha)(x' + y'\alpha) &\equiv xx' + (xy' + x'y)\alpha + yy'(n\alpha - 1) \equiv (xx' - yy') + (xy' + x'y + yy'n)\alpha \pmod{p}. \end{aligned}$$

The identity 1 belongs to F . Also, the additive inverse and the multiplicative inverse for nonzero elements exist since

$$\begin{aligned} -(x + y\alpha) &\equiv (-x) + (-y)\alpha \pmod{p}, \\ (x + y\alpha)^{-1} &\equiv \frac{x + ny}{x^2 + nxy + y^2} + \frac{-y}{x^2 + nxy + y^2}\alpha \pmod{p} \end{aligned}$$

by simple checking (note that $x^2 + nxy + y^2 \neq 0$ since otherwise we can take $t = -xy^{-1}$ so that $n \equiv t + t^{-1} \pmod{p}$). Thus, F is a field with p^2 elements. Also, we have $n = \alpha + \alpha^{-1}$ in this field (from now on, all equality means equality as elements in F). As in the first case, we can use $n^3 - 3n + 1 = 0$ to deduce $\alpha^3 + 1 + \alpha^{-3} = 0$, and hence $\alpha^9 = 1$. Again, let d be the order of α in F . If $d = 1, 3$, then $\alpha^3 = 1$ and we can deduce $p = 3$ as before. If $d = 9$, then $9 \mid p^2 - 1$ since the multiplicative group of F has $p^2 - 1$ elements (all elements in F except 0). This implies $p \equiv \pm 1 \pmod{9}$.

Therefore, $p = 3$ or $p \equiv \pm 1 \pmod{9}$ hold in any case. $\square$